Lab2 Docker

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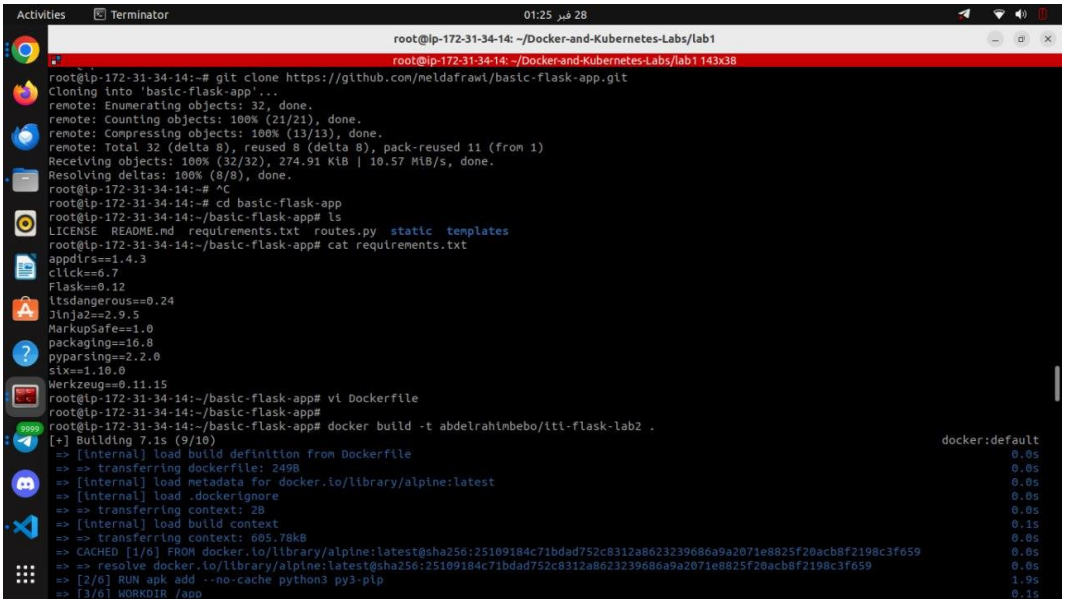
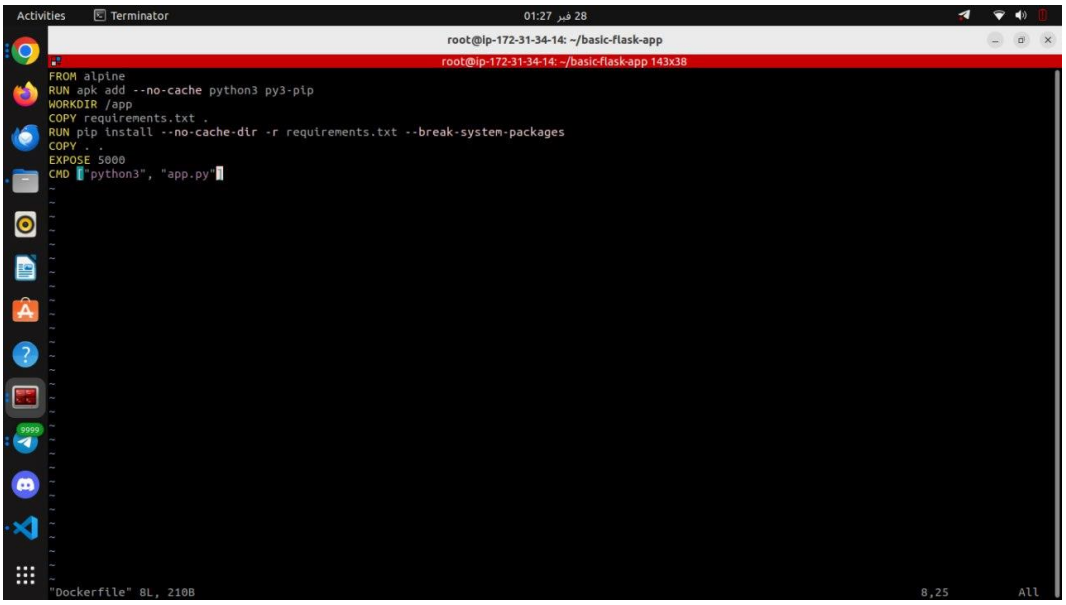
Q1:

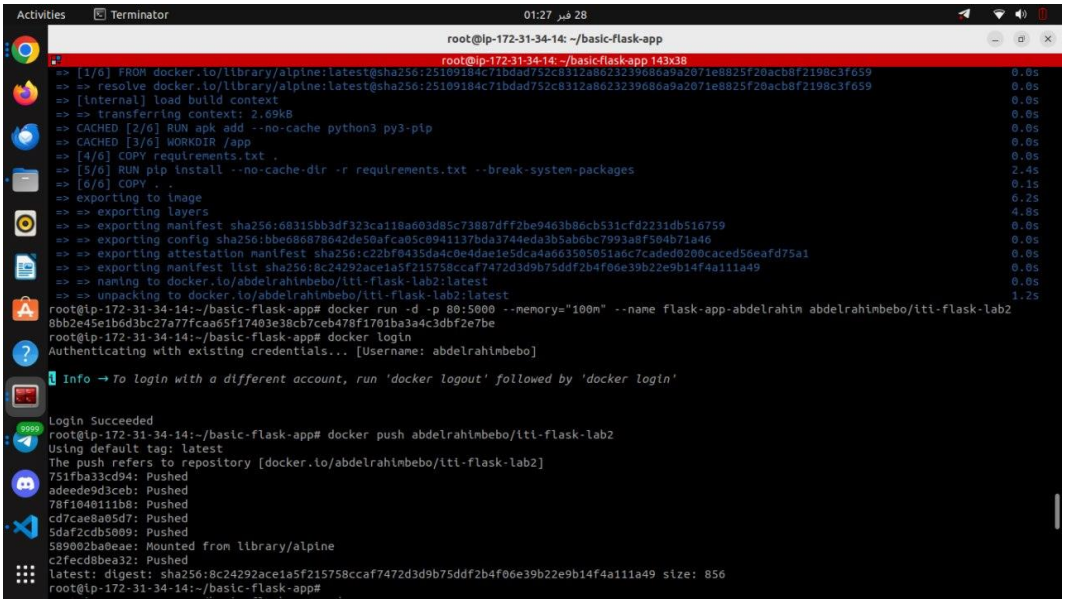
1. Cloning from the github:

a. 

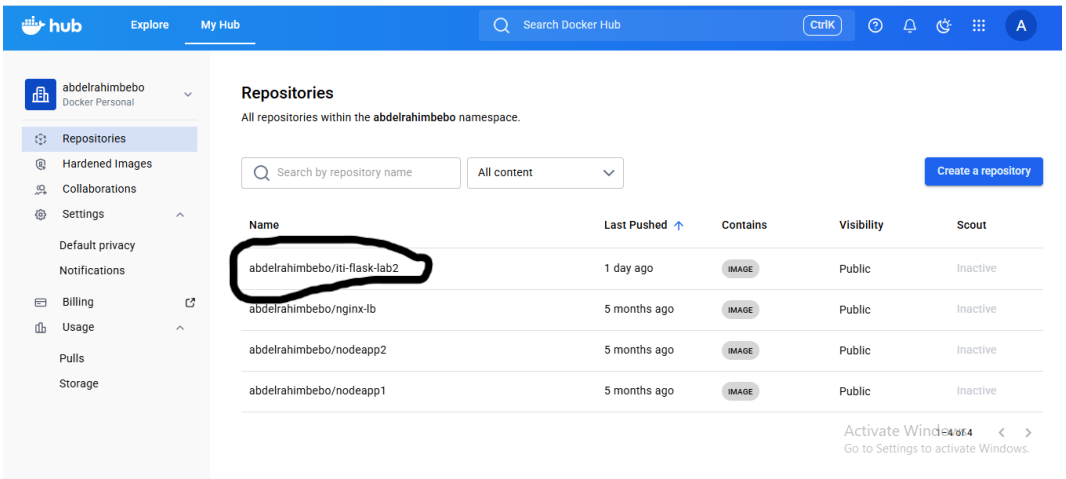
2. Creating the Dockerfile that will be built and create the image from it

a. 

3. Building, Running, and Pushing to Dockerhub:

a. 

4. The Dockerhub after pushing the image:

a. 

Q2:

1. Creating the Network with the specifications in Lab

a.

```
root@ip-172-31-34-14:~# docker network create --subnet 10.0.0.0/8 iti-network
d57d4adbbfb49726d72adeddfc10b840c1337f0dabe735f0fc323f655d5653b9
root@ip-172-31-34-14:~# docker network ls
NETWORK ID     NAME          DRIVER    SCOPE
cdb83be6e312   bridge        bridge    local
d8de63793dca   host          host      local
d57d4adbbfb4   iti-network   bridge    local
7b17fccbcdcb   none          null      local
root@ip-172-31-34-14:~# mkdir html
nano html/index.html
```

2. The editing of the html file to make the output as asked in the lab:

a.

```
root@ip-172-31-34-14: ~
root@ip-172-31-34-14: ~ 127x38
  GNU nano 7.2                    html/index.html
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```

3. The directory of the html file and the command of running the container:

a.

```
root@ip-172-31-34-14:~# mkdir html
nano html/index.html
root@ip-172-31-34-14:~# nano html/index.html
root@ip-172-31-34-14:~# docker run -d --network iti-network --name webserver-iti -p 8080:80 -v $(pwd)/html:/usr/share/nginx/htm
l nginx:alpine
Unable to find image 'nginx:alpine' locally
alpine: Pulling from library/nginx
5e7756927bef: Pull complete
399d0898a94e: Pull complete
6d397a54a185: Pull complete
6b7b6c7061b7: Pull complete
955a8478f9ac: Pull complete
3e2c181db1b0: Pull complete
bca5d04786e1: Pull complete
dda176b4ea00: Download complete
8a29e1f4877b: Download complete
Digest: sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca557a8d7702ba
Status: Downloaded newer image for nginx:alpine
6f4b2e84690a58d88f05dc9547d3f5d4b5b219aea7fd7d2a1522f1c5b52e7c78
root@ip-172-31-34-14:~#
```

4. Testing the container by curl :

a.

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Terminator
16:09 28 فبر
root@ip-172-31-34-14: ~
root@ip-172-31-34-14: ~ 127x38
root@ip-172-31-34-14:~# curl -I localhost:8080
HTTP/1.1 200 OK
Server: nginx/1.29.5
Date: Sat, 28 Feb 2026 14:09:16 GMT
Content-Type: text/html
Content-Length: 37
Last-Modified: Sat, 28 Feb 2026 13:57:14 GMT
Connection: keep-alive
ETag: "69a2f43a-25"
Accept-Ranges: bytes
root@ip-172-31-34-14:~# curl localhost:8080
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root@ip-172-31-34-14:~#